Methodology of Function Analysis [IT: Analisi delle Funzioni]

Methodological framework for analytic analysis and graphical interpretation.

1. Analytic Classification and Domain [IT: Classificazione e Dominio]

Question 1: What is the position of the independent variable x?

- **Under a root sign [IT: Sotto radice]?**
  - $\rightarrow$ *Even index [IT: Indice pari] (e.g., $\sqrt{x}$):* The radicand must be non-negative in $\mathbb{R}$.
    Existence Conditions (E.C.) [IT: C.E.]: Set radicand ≥ 0. (*Irrational Function* [IT: Funzione Irrazionale]).
  - $\rightarrow$ *Odd index [IT: Indice dispari] (e.g., $\sqrt[3]{x}$):* Defined $\forall x \in \mathbb{R}$.
    Action: No restrictions. (*Irrational Function*).
- **In the denominator of a fraction [IT: Al denominatore]?**
  - $\rightarrow$ *Algebraic principle:* Division by zero is undefined.
    E.C. [IT: C.E.]: Set the entire denominator $\neq 0$. (*Rational Fractional Function* [IT: Razionale Fratta]).

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2. Graphical Decoding and Interpretation [IT: Interpretazione del Grafico]

Question 2: What analytic information can be deduced from the graph?

- **Dashed lines (Asymptotes) [IT: Rette tratteggiate / Asintoti]:**
  They indicate boundaries or limits to infinity. A vertical asymptote signals an x-value where the function diverges (e.g., the zeros of a denominator).
- **Filled dot [IT: Punto pieno] (•):**
  The point belongs to the graph. The function is defined at that extremum (common with $\geq$ inequalities in E.C.).
- **Empty dot [IT: Punto vuoto] (○):**
  The point does not belong to the graph. It indicates a discontinuity or a value strictly excluded from the domain.
- **Passing through the origin [IT: Passaggio per l'origine] $(0, 0)$ e intersections:**
  Passing through the origin implies that substituting $x = 0$ yields $y = 0$.

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3. Symmetry Analysis [IT: Analisi delle Simmetrie]

Question 3: Does the function show geometric regularities verifiable algebraically?

- **Algebraic test:** Substitute x with $(-x)$ and compare the result with the original expression.
  - $\rightarrow$ If $f(-x) = f(x)$: The function is **EVEN [IT: PARI]**. (Symmetric with respect to the y-axis).
  - $\rightarrow$ If $f(-x) = -f(x)$: The function is **ODD [IT: DISPARI]**. (Symmetric with respect to the origin $(0, 0)$).

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4. Typological Classification and Invertibility [IT: Classificazione e Invertibilità]

Question 4: What is the nature of the mapping and the criteria for inversion?

- **Injective [IT: Iniettiva] (Uniqueness of values):**
  Distinct elements of the domain map to distinct y-values.
  Graphical test: Any horizontal line intersects the graph at most once.

- **Surjective [IT: Suriettiva] (Codomain coverage):**
  Every element in the target set is reached by the function.
  Graphical test: The projection of the graph fully covers the y-axis.

- **Bijective [IT: Biiettiva]:**
  The function is both injective and surjective (perfect 1-to-1 mapping).

- **Existence of the Inverse Function [IT: Funzione Inversa] (f^{-1}):**
  Reverse calculation is possible **if and only if** the function is bijective.

  → *Algebraic procedure:* Isolate the independent variable x in terms of y. Then, formally swap the variables to restore the standard notation $y = f^{-1}(x)$.

  → *Graphical relation:* The graph of $f^{-1}(x)$ is symmetric to $f(x)$ with respect to the bisector of the first and third quadrants (line $y = x$).